Weak convergence of a fast mean-reverting rough Heston model with jumps to a general class of Lévy processes

Abstract

We show weak convergence of the marginals for a re-scaled rough Heston model to a Normal Inverse Gaussian (NIG) Lévy process. This shows we can obtain such a limit without having to impose that the true Hurst exponent H for the model is $\frac{1}{2}$ as in [AC24],[1] or that $H \searrow -\frac{1}{2}$ as in [AAR25], so the result potentially has increased financial relevance. We also extend to the case when V has jumps, where we show weak convergence of the finite-dimensional distributions of the integrated variance to a deterministic time-change of the first passage time process to lower barriers for a more general class of spectrally positive Lévy processes.[2]

1 Asymptotics for the terminal log stock price for a re-scaled rough Heston model

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{Q})$ throughout with filtration $(\mathcal{F}_t)_{t\geq 0}$ satisfying the usual conditions.

Proposition 1.1 *Consider a re-scaled rough Heston model for a log stock price process X_t^ε:*

$$\begin{aligned} dX_t^\varepsilon &= -\frac{1}{2}V_t^\varepsilon dt + \sqrt{V_t^\varepsilon}dB_t, \\ V_t^\varepsilon &= V_0 + \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\Big(\frac{1}{\varepsilon}\lambda(\theta - V_s^\varepsilon)ds + \frac{1}{\varepsilon}\sigma\sqrt{V_s^\varepsilon}dW_s\Big), \end{aligned} \tag{1}$$

where B and W are two Brownian motions with $dB_tdW_t = \rho dt$ with $\rho \in [-1,0]$, $\alpha \in (\frac{1}{2}, 1)$ and $\lambda, \sigma > 0$. Then (for $t > 0$ fixed) X_t^ε tends weakly to X_t as $\varepsilon \to 0$, where X is a Normal Inverse Gaussian Lévy process which does not depend on $H = \alpha - \frac{1}{2}$.

Proof. Let $I^\alpha(f)(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}f(s)ds$ denote the αth-order fractional integral of a function f for $\alpha \in (0,1]$, and (without loss of generality), we assume $X_0^\varepsilon = 0$. Then for $p \in (0,1)$ (which will be sufficient for our purposes when we invoke the [Bill86] weak convergence result below)

$$\mathbb{E}[e^{pX_t^\varepsilon}] = e^{V_0 I^{1-\alpha}\phi_\varepsilon(t) + \frac{1}{\varepsilon}\lambda\theta I^1\phi_\varepsilon(t)} \tag{2}$$

where ϕ_ε is the unique solution of the non-linear Volterra integral equation (VIE):

$$\phi_\varepsilon(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + \frac{1}{\varepsilon}(\rho p\sigma - \lambda)\phi_\varepsilon(s) + \frac{1}{\varepsilon^2}\frac{1}{2}\sigma^2\phi_\varepsilon(s)^2\Big)ds$$

(see e.g. Section 4 in [ER19] or Section 7 in [ALP19]). Now let $\phi_\varepsilon(t) = \varepsilon\psi(\varepsilon^q t)$. Then

$$\begin{aligned} \varepsilon\psi(\varepsilon^q t) &= \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(\varepsilon^q s) + \frac{1}{2}\sigma^2\psi(\varepsilon^q s)^2\Big)ds \\ &= \frac{1}{\Gamma(\alpha)}\int_0^{\varepsilon^q t} (t-u\varepsilon^{-q})^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(u) + \frac{1}{2}\sigma^2\psi(u)^2\Big)du\,\varepsilon^{-q} \\ &= \frac{\varepsilon^{-q(\alpha-1)}}{\Gamma(\alpha)}\int_0^{\varepsilon^q t} (\varepsilon^q t-u)^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(u) + \frac{1}{2}\sigma^2\psi(u)^2\Big)du\,\varepsilon^{-q} \end{aligned}$$

where we set $\varepsilon^q s = u$ in the second line, so $\varepsilon^q ds = du$. Then setting $\varepsilon^q t \mapsto t$, we see that

$$\varepsilon\psi(t) = \frac{\varepsilon^{-q\alpha}}{\Gamma(\alpha)}\int_0^t (t-u)^{\alpha-1}F(\psi(u))du \tag{3}$$

[1] The H parameter in [AC24] is not fixed to be $\frac{1}{2}$, but the Hurst exponent for their family of V^ε processes is of course $\frac{1}{2}$ since the models are all standard Heston for $\varepsilon > 0$.

[2] Note that for citations to results in [ALP19], [BLP24], [BPS24], [FGS21] and [GK19] we are referring to Lemma or Theorem numbers in the published, not the arXiv versions.

where $F(w) = \frac{1}{2}(p^2 - p) + (\rho p\sigma - \lambda)w + \frac{1}{2}\sigma^2 w^2$. If now we let $q = -\frac{1}{\alpha}$, the VIE (3) is independent of ε, so

$$\phi_\varepsilon(t) = \varepsilon\psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}}).$$

Hence for every $t > 0$ (from Lemma 4.5 in [FGS21])

$$\lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\phi_\varepsilon(t) \;=\; \lim_{t\to\infty}\psi(t) \;=\; \psi(\infty) \;=\; U_1(p) \;:=\; \frac{1}{\sigma^2}[\lambda - p\sigma\rho - \sqrt{\lambda^2 - 2\lambda\rho\sigma p + \sigma^2 p(1-p\bar{\rho}^2)}]$$

for $p \in (0,1)$, where $\psi(\infty)$ is the smallest root of F and $\bar{\rho}^2 = 1-\rho^2$, and the convergence to $\psi(\infty)$ is rapid for $\varepsilon \ll 1$ since $\frac{1}{\varepsilon}\phi_\varepsilon(t) = \psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}})$. More precisely, Lemma 4.5 in [FGS21] can be applied to $-\psi$, which solves the following VIE:

$$-\psi(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-u)^{\alpha-1}F_1(-\psi(u))du, \quad \text{with } F_1(w) = \frac{1}{2}(p-p^2) + (\rho p\sigma - \lambda)w - \frac{1}{2}\sigma^2 w^2$$

on $\mathbb{R}_+$, since $F_1(0) > 0$ and F_1 is analytic and decreasing on $\mathbb{R}_+$, the aforementioned result in [FGS21] implies that $-\psi(t)$ converges as $t \to \infty$ to the positive root of F_1, which coincides with $-U_1(p)$. Then for the exponent in (2), we know that $\frac{1}{\varepsilon}\phi_\varepsilon(.) = \psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}})$ is continuous on $\mathbb{R}_+$ and admits finite limit at ∞, and thus is bounded), so (by the dominated convergence theorem) we see that

$$V_0 I^{1-\alpha}\phi_\varepsilon(t) + \frac{\lambda\theta}{\varepsilon}I^1\phi_\varepsilon(t) \;=\; \frac{V_0}{\Gamma(1-\alpha)}\int_0^t (t-s)^{-\alpha}\phi_\varepsilon(s)ds + \lambda\theta\int_0^t \frac{1}{\varepsilon}\phi_\varepsilon(s)ds \;\to\; 0 + \lambda\theta U_1(p)t$$

as $\varepsilon \to 0$, and $\lambda\theta U_1(p)t$ is the log mgf of a Normal Inverse Gaussian Lévy process (see e.g. Remark 2.3 in [FJ11]). Then from e.g. the solution to Problem 30.4 on Page 573 in [Bill86] (which is also used in [GK19]), X_t^ε tends weakly to the marginal law of an NIG process. ■

1.1 Adding jumps into the V^ε process

If we add a (re-scaled) independent jump component into the V^ε process as in Eq 1 in [BPS24], then (from Theorem 3 in [BPS24], see also slide 6 in [Cuch22]), then $F(\psi(u))$ in (3) has an additional term

$$V_\varepsilon(\varepsilon\psi(u))$$

where $V_\varepsilon(p) := \int(e^{pz} - 1 - p|z|)\nu(\varepsilon dz)$ is the log mgf of the re-scaled Lévy component, so

$$V_\varepsilon(\varepsilon p) \;=\; \int(e^{p\varepsilon z} - 1 - \varepsilon p|z|)\nu(\varepsilon dz) \;=\; \int(e^{px} - 1 - p|x|)\nu(dx) \;=\; V_1(p)\,.$$

In this case $\frac{1}{\varepsilon}\phi_\varepsilon(t)$ tends to the smallest root of $F_J(u) = \frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)u + \frac{1}{2}\sigma^2u^2 + V_1(u)$, assuming a root still exists, and note V_1 (and F_J) are convex (see e.g. Lemma 2.2.5 in [DZ98]).

From here on, we assume $\nu = \nu(z)dz$ only has non-negative support and (state conditions). Then for a spectrally positive Lévy process Z with Lévy triple (b, σ^2, ν) with $b = \rho p\sigma - \lambda$, we know that $\mathbb{E}(e^{uZ_t}) < \infty$ for $u < M$, and (for u in this range) $e^{uZ_t - \Lambda(u)t}$ is an $\mathcal{F}^Z$-martingale.

Let $\Lambda(u) = bu + \frac{1}{2}\sigma^2u^2 + V_1(u)$. Hence from the optional stopping theorem

$$\mathbb{E}(e^{-qH_a}) \;=\; e^{-\Phi(q)a} \;=\; e^{\Phi(q)|a|} \tag{4}$$

for $q \geq 0$ and $a < 0$ where $H_a = \inf\{t \geq 0 : Z_t < a\}$ and $\Phi(q) \leq 0$ is smallest root of $\Lambda(u) = q$ (since we have a *lower* barrier here), see e.g. Eq 2.5 in [KKR13] for details and the limiting argument using the dominated convergence theorem [3], and (4) also holds for $q < 0$ if q is greater than the minimum value attained by $\Lambda(.)$.

If we now let $W_t^{(-\frac{1}{2})} = W_t^\perp - \frac{1}{2}t$ where $W^\perp$ is another Brownian motion independent of Z and everything else, then for p in some non-zero interval around zero, we see that

$$\mathbb{E}(e^{pW_{H_{-t}}^{(-\frac{1}{2})}}) \;=\; \mathbb{E}(e^{\frac{1}{2}(p^2-p)H_{-t}}) \;=\; e^{t\Phi_p(-\frac{1}{2}(p^2-p))} \tag{5}$$

and $\Phi_p(-\frac{1}{2}(p^2-p))$ is the smallest root of F_J (note that H_{-t} also depends on p here since $b = \rho p\sigma - \lambda$).

(5) is the mgf of $\eta_{\tau_t^-}$, where $\eta_t = W_t - \frac{1}{2}t$ and $\tau_t^- = \inf\{s : \xi_s < -t\}$, where $\xi_t = -\lambda t + \sigma(\rho W_t + \bar{\rho}\tilde{W}_t) + J_t$, where W and $\tilde{W}$ are independent and J is a spectrally positive Lévy process (independent of $(W, \tilde{W})$), with $\log\mathbb{E}(e^{uJ_1}) = V_1(u)$.

[3] in [KKR13] they take the largest root since they are considering an upper barrier

To see this, we note that the joint mgf of (ξ_1, η_1) is

$$\Psi(u,p) \quad = \quad \log \mathbb{E}(e^{u\xi_1 + p\eta_1}) \quad = \quad \Lambda_p(u) + \frac{1}{2}(p^2 - p)$$

with $\Lambda_p(u) = -\lambda u + \frac{1}{2}\sigma^2 u^2 + V_1(u) + \rho\sigma p u$. Now define $\kappa(p)$ by the branch through zero:

$$\Lambda_p(\kappa(p)) + \frac{1}{2}(p^2 - p) \quad = \quad 0$$

with $\kappa(0) = 0$. Then $\kappa(p) = \Phi_p(-\frac{1}{2}(p^2 - p))$, and

$$\mathbb{E}(e^{\kappa(p)\xi_{\tau_t^-} + p\eta_{\tau_t^-}}) \quad = \quad \mathbb{E}(e^{-\kappa(p)t + p\eta_{\tau_t^-}}) \quad = \quad 1$$

so

$$\mathbb{E}(e^{p\eta_{\tau_t^-}}) \quad = \quad e^{\kappa(p)t}$$

assuming $\tau_t^- < \infty$ a.s., which is true if $\Lambda_0'(0+) \leq 0$, where $\Lambda_0(u) = -\lambda u + \frac{1}{2}\sigma^2 u^2 + V_1(u)$ and $V_1'(0) < \infty$.

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